

JAMES RUSE AGRICULTURAL HIGH SCHOOL

Student ID:



PHYSICS

HIGHER SCHOOL CERTIFICATE

2004

TRIAL EXAMINATION

General Instructions

Reading time – 5 minutes

Working time – 3 hours

Board-approved calculators may be used

Write using blue or black pen

Draw diagrams using pencil

A Data Sheet, Formulae Sheets and a Periodic Table are provided.

Write your Student Number in the space provided.

Section I	Theory Section:	Total Marks (75)	Contains Part A and Part B
	Part A	15 marks,	allow about 30 mins
	Part B	60 marks,	allow about 1 hour 45 min

Section II	Option Section:	Total marks (25),	allow about 45 mins
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Section I Part A: Answer Space. Place a cross in the correct space.

	A	B	C	D
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Section I Part A:

Answer all questions in the multiple choice answer space.

1. A satellite of the earth (radius 6×10^6 m) takes 96 minutes to orbit the earth at an altitude of 200 km above the earth's surface. Using this data, the mass of the earth is closest to:-
 - a) 6.0×10^{24} kg
 - b) 5.8×10^{24} kg
 - c) 5.2×10^{24} kg
 - d) 4.3×10^{24} kg

2. The velocity of a particle 2 seconds after it has been projected horizontally at 30 ms^{-1} in the earth's field is approximately:-
 - a) 20 ms^{-1}
 - b) 30 ms^{-1}
 - c) 36 ms^{-1}
 - d) 42 ms^{-1}

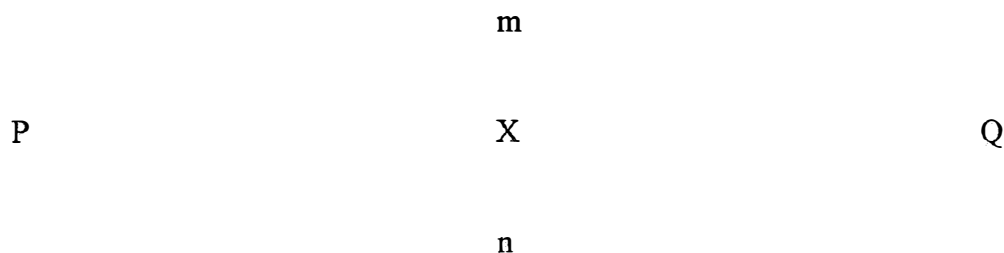
3. If $g(\text{moon}) = 1.6 \text{ ms}^{-2}$ and $g(\text{mars}) = 3.7 \text{ ms}^{-2}$ the ratio of the forces experienced by a 30 kg object when taken to the moon then mars is closest to:-
 - a) 1.6 : 3.7
 - b) 48 : 3.7
 - c) 1.6 : 117
 - d) 1 : 1

4. The gravitational attraction between two 5 gram masses separated by a distance of 15 cm is closest to:-
 - a) $G \div 9$
 - b) $G \times 10^{-2} \div 9$
 - c) $G \times 10^{-6} \div 9$
 - d) $G \times 10^{-1} \div 45$, where "G" is the universal gravitational constant.

5. The period "T" of a simple pendulum is given by $T = 2\pi\sqrt{l \div g}$, where "l" is the length and "g" is the earth's gravity constant. Which of the following would produce a straight line graph passing through the origin?
 - a) T vs $\sqrt{l}$
 - b) T vs l
 - c) T vs $\sqrt{g}$
 - d) T vs g

6. Torque may be described as:-
 - a) the turning moment of a force
 - b) force times distance (perpendicular to the force)
 - c) force times distance (parallel to the force)
 - d) two of the above

7. Energy losses occur as energy is fed through transmission lines from the generator to the consumer. These losses are minimised in practice by:-
- using transmission lines of the lowest possible resistance value.
 - sending the energy in high current/low voltage form
 - sending the energy in high voltage/low current form
 - two of the above
8. A rectangular loop of wire conductor is rotated inside a uniform magnetic in the usual orientation to produce an emf. The emf is proportional to:-
- the rate of change of magnetic field strength
 - the rate of change of area as the loop rotates
 - the rate of change of flux
 - two of the above
9. An “ideal” transformer has 200 turns in the primary winding and 40 000 turns in the secondary. If a current of 2 mA flows in the secondary and the power output is 100 watts the primary voltage is:-
- 200 V
 - 250 V
 - 400 V
 - 50 000 V
10. In an electric meter in which the magnetic pole pieces are curved to produce a radial magnetic field:-
- the scale is always linear
 - the rectangular conducting loop is always parallel to the magnetic field
 - the rectangular conducting loop is always perpendicular to the magnetic field
 - two of the above
11. Oppositely charged objects P and Q are separated as in the diagram below.



A third charge X is placed and held half way between P and Q. To place X in a region of higher electric field intensity X could be moved towards:-

- m or n
- P or Q
- P only
- Q only

12. A charged particle may be able to move in a straight line through a region containing an electric and a magnetic field if it enters:-
- a) the magnetic field before the electric field
 - b) the electric field before the magnetic field
 - c) a region where the magnetic and electric fields are perpendicular to each other.
 - d) a region where the magnetic and electric fields are parallel with each other
13. A photon of wavelength 400 nm would have energy of approximately:-
- a) 3.1 eV
 - b) 4.9 eV
 - c) 2.4×10^{-40} J
 - d) 9.0×10^{-19} J
14. A monochromatic light source is causing electrons to be emitted from a metal surface. Greatly increasing the intensity of the light source will result in:-
- a) electrons being emitted with greater energy
 - b) many more electrons being emitted
 - c) slightly more electrons being emitted
 - d) no change in the number of emitted electrons
15. The number of free electrons that can drift from atom to atom is least in:-
- a) insulators
 - b) p type semiconductors
 - c) n type semiconductors
 - d) conductors

End Section 1 Part A multiple choice

Student ID:

Section I Part B: Questions 16-29

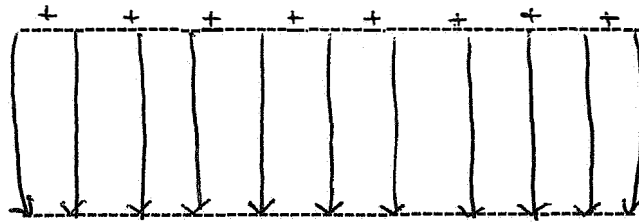
Answer all questions in the space provided.

Show all relevant working .

16. At lift-off a rocket has a mass of 35,000 kg, of which 80% is fuel. The rocket takes off vertically and has a thrust of 400,000 N. Just before the fuel runs out the rocket is accelerating vertically. Assuming that the thrust is constant find:-
a) the acceleration at lift-off. (2 marks)
- b) the g-force on the rocket just before the fuel runs out. (2 marks)
17. Compare qualitatively low earth and geo-stationary orbits. (4 marks)
18. Explain how the production and reception of radio waves may be demonstrated in the school laboratory. (4 marks)

Student ID:

19. A pair of parallel conducting plates is separated by a distance of 20mm, as in the diagram below. The potential difference across the plates is 4000 volts. The upper plate is positive.



- a) Draw the electric field lines on the diagram using all the usual conventions. (1 mark)
- b) A charge of $+ 1.6 \times 10^{-19}$ C is placed 5mm from the negative plate. Find the force acting on the charge. (disregard gravity) (2 marks)
- c) The charge is now moved to a position 5mm from the positive plate. What force now acts on the charge? Give a reason for your answer. (2 marks)
20. Describe how “doping” a semiconductor can change its electrical properties. (3 marks)

Student ID:

21. a) Identify a practising male or female Australian scientist. (1 mark)
- b) In what area is he/she currently working? (1 mark)
- c) Give information about his/her research. (2 marks)
22. What is the nature of an inertial frame of reference? (2 marks)
23. a) Spaceship “Interstellar” is travelling through space at a speed at which the time passed on the spaceship in one earth day appears to take only 10 hours. At what fraction of the speed of light is the spaceship travelling? (3 marks)
- b) If the “Interstellar” is 25m long with respect to an observer who is onboard, what will an observer who is stationary with respect to the spaceship measure its length to be as it passes at the speed in a) above? (2 marks)

Student ID:

24. The table below shows the radius of orbit for a number of planets in our solar system. For this question assume that the orbits are circular and the planets move at a constant speed.

Planet:	Mercury	Venus	Earth	Jupiter	Saturn
radius of orbit ($\times 10^6$ km)	58.5	109	150	780	1430

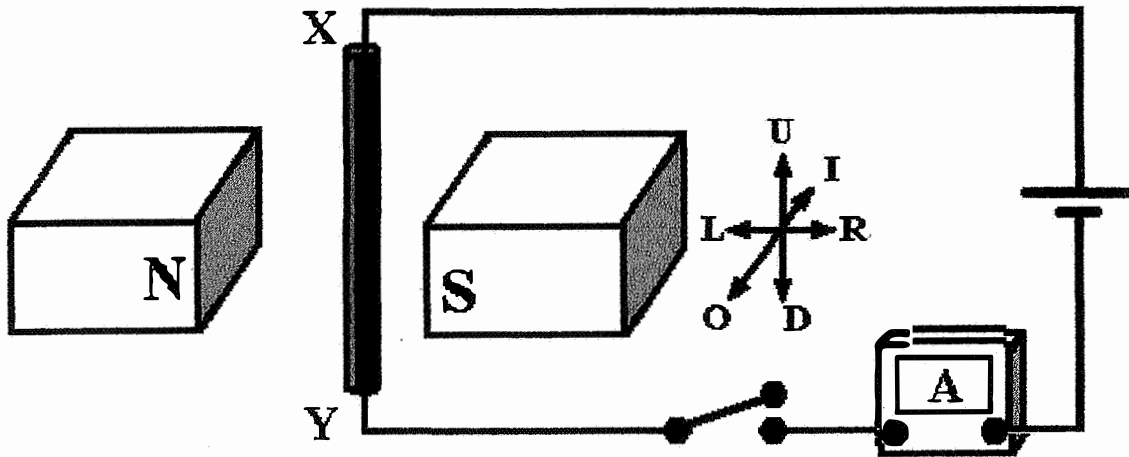
- a) Calculate the time for Jupiter to complete one orbit in terms of earth years (2 marks)

- b) Calculate the orbital speed of Jupiter in ms^{-1} . (2 marks)

- c) If the mass of Jupiter is 1.9×10^{27} kg find the force the sun exerts on it to keep it in orbit. (2 marks)

Student ID:

25. In the diagram below two bar magnets are producing a uniform magnetic field of 0.4 Tesla. The conductor X-Y is 5.0 cm long and has a total resistance of 10.0 ohms, distributed evenly along the conductor. Only 3.0 cm of the conductor is actually inside the magnetic field. The battery voltage is 100.0 volts.



A perpendicular axis system is shown in the diagram, labelled: up (U), down (D), left (L), right (R), into page (I) and out of page (O). The conductor is free to move in any of these directions.

- a) Calculate the current flowing in the circuit just as the switch is closed, before the conductor starts to move. (1 mark)
- b) Calculate the magnitude of the magnetic force exerted on the conductor after the switch is closed. (1 mark)
- c) State the direction: U, D, R, L, I or O, in which the conductor will start to move. (1 mark)

Student ID:

25 cont. d) As the conductor starts to move in response to this magnetic force, the reading on the ammeter will change. State how the reading will change and explain the reasoning behind your answer. (3 marks)

26. A particle of charge $3.2 \times 10^{-19} \text{ C}$ and mass 4 atomic mass units (u) enters a uniform magnetic field of strength 0.2 Tesla with a velocity of $4 \times 10^6 \text{ ms}^{-1}$ whereupon it moves in a circular path. Assuming it completes a semicircle, then leaves the field, determine the time it spends in the field. (4 marks)

Student ID:

27. Explain the particle theory of light in terms of photons with particular energy and frequency. (5 marks)

- 28 a) Explain the role of transformers in electricity substations (3 marks)

Student ID:

28 cont.

- b) Why do some electrical appliances in the home use a transformer? Give an example.
(3 marks)

- 29 Discuss the effect on the magnitude of the force on a current-carrying conductor when the angle between the direction of the external magnetic field and the direction of the length of the conductor is varied.
(2 marks)

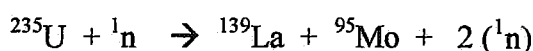
End of PART B

Student ID:

Section II Option: Quanta to Quarks.
provided. Show all working.

Answer all questions in the writing booklet

- 1 What is meant by the term “transmutation”. (1 mark)
- 2 Outline how you performed a first hand investigation to observe the hydrogen spectrum. (4 marks)
- 3 Account for the need for the strong nuclear force and describe its properties. (4)
4. One atom of U 235 can fission to produce La 139 and Mo 95 and two neutrons as in the equation below:- (do not consider electrons in any calculations)



where La 139 = 138.8061 amu

Mo 95 = 94.9057 amu

U 235 = 235.0439 amu

n = 1.0087 amu

- a) Using the information above calculate the mass defect of U 235 in amu. (2 marks)
 - b) How many joules is this equivalent to? (2 marks)
 - c) If 235 grams of U contains 6×10^{23} atoms how many joules of energy would be released by the complete conversion of one kg of U 235 into energy? (3 marks)
5. Protons in the nucleus attract each other gravitationally and repel electrically. Evaluate the relative contributions of the two forces. (Numerical answer required.). Given the electrical force is kQq/R^2 , where k is the coulomb constant 9×10^9 SI units, and charges Q and q coulombs are separated by R metres. (3 marks)
- 6 Explain the concept of mass defect and indicate how this relates to the release of energy in a fission reaction. (6 marks)

End of Section II Option

----- END OF TEST -----

JAMES RUSE AGRICULTURAL HIGH SCHOOL

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2004

TRIAL EXAMINATION SOLUTIONS

General Instructions

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Section I Part A: **Answer Space.** Place a cross in the correct space.

	A	B	C	D
1				×
2			×	
3	×			
4		×		
5	×			
6				×
7			×	
8		.		×
9		×		
10				×
11		×		
12			×	
13	×			
14		×		
15	×			

Section I Part B: Questions 16-29

Answer all questions in the space provided.

Show all relevant working.

16. At lift-off a rocket has a mass of 35,000 kg, of which 80% is fuel. The rocket takes off vertically and has a thrust of 400,000 N. Just before the fuel runs out the rocket is accelerating vertically. Assuming that the thrust is constant find:-

a) the acceleration at lift-off.

(2 marks)

$$T - mg = ma$$

$$\text{mass of rocket} = 35000 - 28000 = 7000 \text{ kg}$$

$$T = 400,000$$

$$a = \frac{T - mg}{m} = \frac{400000 - 49000}{7000} = 46.2 \text{ ms}^{-2}$$

b) the g-force on the rocket just before the fuel runs out.

(2 marks)

$$\text{after fuel burn, mass of rocket} = 35000 - 28000 = 7000 \text{ kg}$$

$$T = 400,000 \quad m = 7000 \quad a = \frac{T}{m} = 47.1 \text{ ms}^{-2}$$

$$g = \frac{47.1 + 9.8}{9.8} = 5.8 \text{ g}$$

$$a = \frac{T - mg}{m} = \frac{400000 - (7000 \times 9.8)}{7000} = 47.34$$

17. Compare qualitatively low earth and geo-stationary orbits.

(4 marks)

Low earth	Geostationary
lies on outer rim of atmosphere	lies near upper edge of V.A. belt
about 250 - 1000 km high	altitude 38000 km
period of 90 minutes	period about 24 hours
orbital vel. of 28000 km h ⁻¹	orbital speed is 11000 km h ⁻¹
subjected to friction	not subjected to friction

18. Explain how the production and reception of radio waves may be demonstrated in the school laboratory.

(4 marks)

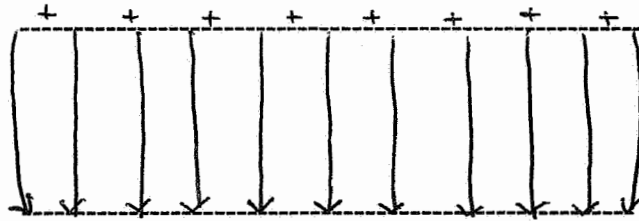
Equipment: High voltage source (induction coil), small portable radio.

Adjust induction coil to produce at a 5 mm spark.

Scan across the AM dial to pick up the radio waves produced by sparking.

or Set up CRO with unbreakable wire attached to input. Place CRO 1.5 m in front of induction coil and observe pattern of wave on screen.

19. A pair of parallel conducting plates is separated by a distance of 20mm, as in the diagram below. The potential difference across the plates is 4000 volts. The upper plate is positive.



- a) Draw the electric field lines on the diagram using all the usual conventions. (1 mark)
- b) A charge of $+1.6 \times 10^{-19}$ C is placed 5mm from the negative plate. Find the force acting on the charge. (disregard gravity) (2 marks)

$$E = \frac{V}{d} = \frac{4000}{20 \times 10^{-3}} = 200000 \text{ V/m}$$

$$F = Eq = 3.2 \times 10^{-14} \text{ N downwards or toward - plate}$$

- c) The charge is now moved to a position 5mm from the positive plate. What force now acts on the charge? Give a reason for your answer. (2 marks)

Same force (3.2×10^{-14} N)
 Electric field is uniform.

20. Describe how "doping" a semiconductor can change its electrical properties. (3 marks)

Doping is the addition of an impurity to a semiconductor. Doping increases the electrical conductivity of the semiconductor.

N-type semiconductors: group 5 element used for doping. The extra e^- is in the conduction band and is mobile, increasing the conductivity. (eg. P or As).

P-type semiconductor: group 3 element (eg. B or Ga) used for doping. As there are only 3 valence e^- , a hole is incorporated, which can be filled by an e^- . The hole facilitates the movement of e^- , i.e. increasing its conductivity.

21. a) Identify a practising male or female Australian scientist. (1 mark)
-
- b) In what area is he/she currently working? (1 mark)
-
- c) Give information about his/her research. (2 marks)
-
-
-
-

22. What is the nature of an inertial frame of reference? (2 marks)

It is a non-accelerating frame of reference (is at rest or moving at constant velocity)
 : a frame of reference where all the laws of motion
 are valid

23. a) Spaceship "Interstellar" is travelling through space at a speed at which the time passed on the spaceship in one earth day appears to take only 10 hours. At what fraction of the speed of light is the spaceship travelling? (3 marks)

$$t_v = \frac{t_o}{\sqrt{1 - \frac{v^2}{c^2}}} \quad t_o = 10 \quad t_v =$$

$$\sqrt{1 - \left(\frac{v}{c}\right)^2} = \frac{t_o}{t_v}$$

$$\frac{v}{c} = \sqrt{1 - \left(\frac{10}{24}\right)^2}$$

$$\frac{v}{c} = 0.9091 \quad v = 0.9091 c \quad \text{or} \quad 0.91 c$$

- b) If the "Interstellar" is 25m long with respect to an observer who is onboard, what will an observer who is stationary with respect to the spaceship measure its length to be as it passes at the speed in a) above? (2 marks)

$$l_v = l_o \sqrt{1 - \frac{v^2}{c^2}}$$

$$= 25 \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

$$= 10.42 \text{ m}$$

24. The table below shows the radius of orbit for a number of planets in our solar system. For this question assume that the orbits are circular and the planets move at a constant speed.

Planet:	Mercury	Venus	Earth	Jupiter	Saturn
radius of orbit ($\times 10^6$ km)	58.5	109	150	780	1430

- a) Calculate the time for Jupiter to complete one orbit in terms of earth years (2 marks)

$$\frac{R^3}{T^2} = \frac{R^3}{T^2}$$

$$T = 11.86 \text{ Earth Years}$$

- b) Calculate the orbital speed of Jupiter in ms^{-1} . (2 marks)

$$v = \frac{2\pi r}{T}$$

$$v = 13105.78 \text{ ms}^{-1} \text{ or } 1.31 \times 10^4 \text{ ms}^{-1}$$

- c) If the mass of Jupiter is 1.9×10^{27} kg find the force the sun exerts on it to keep it in orbit. (2 marks)

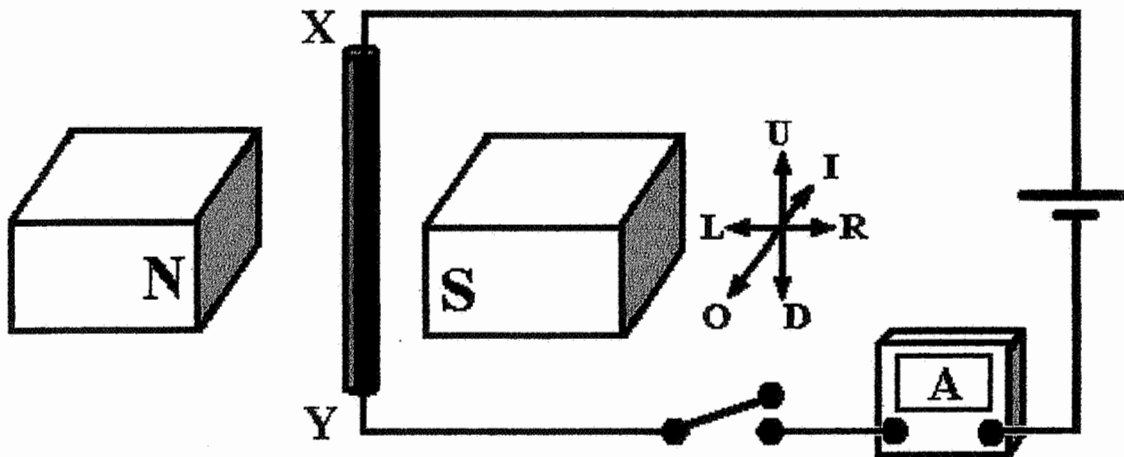
$$F_c = \frac{mv^2}{r}$$

$$= (1.9 \times 10^{27}) (13105.78)^2$$

$$780 \times 10^9$$

$$= 4.18 \times 10^{23} \text{ N towards Sun}$$

25. In the diagram below two bar magnets are producing a uniform magnetic field of 0.4 Tesla. The conductor X-Y is 5.0 cm long and has a total resistance of 10.0 ohms, distributed evenly along the conductor. Only 3.0 cm of the conductor is actually inside the magnetic field. The battery voltage is 100.0 volts.



A perpendicular axis system is shown in the diagram, labelled: up (U), down (D), left (L), right (R), into page (I) and out of page (O). The conductor is free to move in any of these directions.

- a) Calculate the current flowing in the circuit just as the switch is closed, before the conductor starts to move. (1 mark)

$$\frac{I}{R} = \frac{V}{R} = \frac{100}{10} = 10 \text{ A}$$

- b) Calculate the magnitude of the magnetic force exerted on the conductor after the switch is closed. (1 mark)

$$F = BIL$$

$$= 0.4 \times 10 \times 3 \times 10^{-2}$$

$$= 1.2 \times 10^{-1} \quad F = 0.12 \text{ N}$$

- c) State the direction: U, D, R, L, I or O, in which the conductor will start to move.

(1 mark)

O

- 25 cont. d) As the conductor starts to move in response to this magnetic force, the reading on the ammeter will change. State how the reading will change and explain the reasoning behind your answer. (3 marks)

Current will decrease because as the conductor moves towards O the force on the charges in x.y will be upwards (u) direction, opposing the original current in current in direction x to y will decrease (This is an e.g. of back emf)

26. A particle of charge $3.2 \times 10^{-19} \text{ C}$ and mass 4 atomic mass units (u) enters a uniform magnetic field of strength 0.2 Tesla with a velocity of $4 \times 10^6 \text{ ms}^{-1}$ whereupon it moves in a circular path. Assuming it completes a semicircle, then leaves the field, determine the time it spends in the field. (4 marks)

$$F_B = qvB \quad F_c = \frac{mv^2}{r} \quad T = \frac{2\pi r}{v}$$

$$\frac{mv^2}{r} = qvB \quad v = \frac{2\pi r}{T}$$

$$qB = \frac{m \cdot 2\pi r}{T}$$

$$T = \frac{m \cdot 2\pi r}{qB} = 6.50 \times 10^{-9} \text{ s}$$

$$\text{or } \frac{1}{2} T = 3.25 \text{ ns}$$

27. Explain the particle theory of light in terms of photons with particular energy and frequency. (5 marks)

Light can be considered to be packets of energy.
 A photon is the smallest amount of energy a particular frequency of light can have.
 The energy of a photon is given by $E = hf$.
 Energy can be transferred to matter from light in photons.
 However, a photon cannot transfer parts of its energy, but all or none of it.
 The intensity of light is dependent on the number of photons in a given area.
 All photons, regardless of their frequency, have zero rest mass and travel at the speed of light.

28. a) Explain the role of transformers in electricity substations (3 marks)

In transmission substations, transformers increase the voltage & reduce the current when transmitting power for a large distance from ~~where~~ power station to where consumers are located.
 Because current is reduced, energy loss is reduced.
 $P = I^2 R$

In suburban powerstations and in city areas, the voltage is stepped down, so to be safely used at homes, offices etc.

28 cont.

- b) Why do some electrical appliances in the home use a transformer? Give an example. (3 marks)

Homes are provided with 240 V.
 Most electronic circuits are designed to operate
 between 3 - 12 V.
 $\therefore$ Step down transformers are used eg. CD players.
 Other devices, TV & monitors require high voltage to
 operate C.R.T., $\therefore$ step-up transformers are built in.

- 29 Discuss the effect on the magnitude of the force on a current-carrying conductor when the angle between the direction of the external magnetic field and the direction of the length of the conductor is varied. (2 marks)

When conductor is perpendicular to B , the
 force is max.
 When cond. is parallel to $\vec{B}$, the force is zero.
 When conductor is at θ to $\vec{B}$, the force
 is proportional to $\sin \theta$.

End of PART B

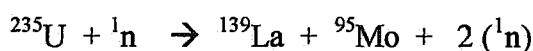
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Section II Option: Quanta to Quarks.
provided. Show all working.

Answer all questions in the writing booklet

a RA element emitting α or β , producing a new element.

- 1 What is meant by the term "transmutation". (1 mark)
- 2 Outline how you performed a first hand investigation to observe the hydrogen spectrum. *H spectral tube, high V source, spectroscope, in the dark* (4 marks)
- 3 Account for the need for the strong nuclear force and describe its properties. *F_g not strong, F_{el} repulsion, weaker. 4 prop - 2 m* (4 marks)
4. One atom of U 235 can fission to produce La 139 and Mo 95 and two neutrons as in the equation below:- (do not consider electrons in any calculations)



where La 139 = 138.8061 amu

Mo 95 = 94.9057 amu

U 235 = 235.0439 amu

n = 1.0087 amu

- a) Using the information above calculate the mass defect of U 235 in amu. *0.3234 amu* (2 marks)
- b) How many joules is this equivalent to? *$4.8 \times 10^{-11} \text{ J}$* (2 marks)
- c) If 235 grams of U contains 6×10^{23} atoms how many joules of energy would be released by the complete conversion of one kg of U 235 into energy? *$1.2 \times 10^{14} \text{ J}$ or $9 \times 10^{16} \text{ J}$* (3 marks)

5. Protons in the nucleus attract each other gravitationally and repel electrically. Evaluate the relative contributions of the two forces. (Numerical answer required.) Given the electrical force is kQq/R^2 , where k is the coulomb constant 9×10^9 SI units, and charges Q and q coulombs are separated by R metres. *$F_{el} = 1.4 \times 10^{36}$ times the F_g*
All working to be shown. (3 marks)

- 6 Explain the concept of mass defect and indicate how this relates to the release of energy in a fission reaction. (6 marks)

mass defect - definition.

$$E = mc^2$$

Fission example.

----- end section II Option -----

----- END OF TEST -----